

Practical 2 Analysis

Searching Algorithm

Searching Algorithms - Short Notes

1. Linear Search

Definition

Linear Search is the simplest searching algorithm that checks each element of the array sequentially until the target element is found or the entire array has been searched.

Key Points :-

- Sequential searching algorithm.
- Checks elements one by one from the beginning.
- Does **not require** the array to be sorted.
- Easy to understand and implement.
- Suitable for small or unsorted datasets.
- Inefficient for large datasets.

Complexity :-

Case	Time Complexity
Best	$O(1)$
Average	$O(n)$
Worst	$O(n)$
Space	$O(1)$

Output :-

```
Enter the size of array: 5
Enter the elements of array : 32
53
22
34
4
Enter element for searching :- 34

><=====><
1.Linear Search
2.Binary Search
3.Exit
Enter Your choice (1-3) :- 1
Element found at Index : 3
```

2. Binary Search

Definition :-

Binary Search is an efficient searching algorithm that works by repeatedly dividing the search interval into two halves until the target element is found.

Key Points :-

- Divide and Conquer algorithm.
- Works **only on sorted arrays**.
- Repeatedly compares the target with the middle element.
- Eliminates half of the search space in every iteration.
- Much faster than Linear Search for large datasets.
- Can be implemented using iteration or recursion.

Complexity :-

Case	Time Complexity
Best	O(1)
Average	O(log n)
Worst	O(log n)
Space (Iterative)	O(1)
Space (Recursive)	O(log n)

Output :-

```
Enter the size of array: 4
Enter the elements of array : 32
53
67
89
Enter element for searching :- 53

><=====><
1.Linear Search
2.Binary Search
3.Exit
Enter Your choice (1-3) :- 2
Element found at Index : 1
Enter the size of array: █
```

Comparison Table

Feature	Linear Search	Binary Search
Technique	Sequential Search	Divide & Conquer
Data Requirement	Sorted or Unsorted	Sorted Array Required
Best Time	$O(1)$	$O(1)$
Average Time	$O(n)$	$O(\log n)$
Worst Time	$O(n)$	$O(\log n)$
Space Complexity (Iterative)	$O(1)$	$O(1)$
Space Complexity (Recursive)	$O(1)$	$O(\log n)$
Easy to Implement	✔ Yes	✔ Yes
Suitable for Large Datasets	✘ No	✔ Yes
Suitable for Unsorted Data	✔ Yes	✘ No

Feature	Linear Search	Binary Search
Array Required	Sorted or Unsorted	Sorted Only
Speed	Slow	Fast
Best Use Case	Small or Unsorted Arrays	Large Sorted Arrays
Best Time	$O(1)$	$O(1)$
Average Time	$O(n)$	$O(\log n)$
Worst Time	$O(n)$	$O(\log n)$
Space	$O(1)$	$O(1)$ (Iterative)